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Electron Microscopy STP

Protocol name: RP03 Preparation of potassium permanganate
Version: 1.1
Status: Approved
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Keywords: Post-staining
Abbreviations: KMN_4 – Potassium Permanganate

Health & Safety requirements

Ensure you are familiar with all relevant local Health and Safety documents and procedures (Material Safety Data Sheets, Procedural Risk Assessments, Standard Operating Procedures etc.) before starting.

Safety critical: All procedures designated safety critical must be performed in a fume hood and the following PPE must be worn - lab coat, gloves, and eye protection.

Procedural Risk Assessments: Handling of potassium permanganate

Materials

- [ ] 15 ml glass scintillation vials with caps
- [ ] Aluminium foil
- [ ] Parafilm .... Agar Scientific AGG3398

Chemicals

- [ ] ddH_2O; double distilled water
- [ ] Potassium Permanganate ($KMNO_4$) .... Merck 1.05082

Procedure

Note: "Potassium Permanganate is used in electron microscopy not only as a fixative but also as a stain. It forms insoluble electron-dense reduction products by reaction with tissue constituents, thus giving rise to image contrast by electron scattering" [1].

Preparation of 1% potassium permanganate

1. Zero a balance with an empty 15 ml glass vial.
2. Weigh 0.1 g of $KMNO_4$ powder into a 15 mL glass vial.
3. Add 10 ml ddH_2O.
4. Close the cap and mix the solution well.
5. Seal with parafilm to prevent oxidisation.

6. Wrap the tube in aluminium foil to protect from light.

7. Store at room temperature.

Potassium Permanganate	State: Solid	Hazard codes: H272, H302, H314, H318, H361d, H400, H410	

References:

[1] Hyatt, M.A. (2000) *Principles and Techniques of Electron Microscopy Biological Applications (Fourth Edition)*. Cambridge University Press. 315-316.